



UNIVERSITY OF CALOOCAN CITY

Computer Studies Department

Bachelor of Science in Information System



Proposed Title:

TaskTap: Local Task and Short-Term Job Management System

Project Description:

Individuals who require help with small tasks like cleaning, tutoring, shopping, repairs, and pet care may not be able to find someone who can help them and who can be trusted in their community. Meanwhile, students, freelancers, or residents seeking a quick cash payout may not have an accessible platform to find local jobs that they can perform and can fit their schedule.

TaskTap is a web-based local task and short-term job management system that connects users within the same area or barangay. Users can create profiles, post tasks, browse available jobs, submit requests, communicate with other users, manage task completion, and record payment information. The application will be developed using C# and ASP.NET Core, with MySQL and Entity Framework Core for data management. A map interface may use Leaflet and OpenStreetMap to display approximate task locations.

TaskTap will be a convenient way for users to discover or submit short-term work in their community. It will help task posters find potential helpers while giving users opportunities to earn through tasks that match their skills. Profiles, ratings, task history, and user reporting will also help promote accountability and trust among users.

General Objectives:

The primary objective of TaskTap is to develop a web-based task and short-term job management system that connects users within the same local area and provides a convenient platform for posting, discovering, requesting, managing, and reviewing small tasks.

The system's purpose is to facilitate access to short-term work available in the local community, and to offer features for organized task management, accountability of users and safer interactions between those who post tasks and those who assist with them.



UNIVERSITY OF CALOOCAN CITY

Computer Studies Department

Bachelor of Science in Information System



Specific Objectives:

Core Technical Features

- To implement secure user authentication and authorization using C# and ASP.NET Core.
- To implement role-based access control for Admin, Standard User, and Guest users.
- To implement user profile management containing basic information, skills, task history, and ratings.
- To implement database management using MySQL and Entity Framework Core.
- To implement soft deletion and account/task archiving for appropriate system records.

Primary Functional Requirements

- To allow users to create and manage task postings containing the task title, category, description, price, location, and deadline.
- To allow users to browse available tasks within their selected local area or barangay.
- To provide search and filtering based on task category, location, price, and deadline.
- To allow users to submit requests to perform available tasks.
- To allow task posters to review, accept, or reject task requests.
- To provide task status management, including available, accepted, in progress, completed, cancelled, and archived statuses.
- To provide an in-app messaging feature that allows users involved in a task to communicate without directly sharing personal contact information.
- To allow users to provide ratings and reviews after a task has been completed.
- To provide notifications for relevant task activities, requests, payment updates, acceptances, messages, and reminders.
- To allow users to report inappropriate tasks or other users for administrative review.
- To provide administrators with tools for managing users, tasks, categories, reports, and archived records.
- To provide an optional map-based view of task locations using approximate locations.
- To record relevant payment information, including payment reference, amount, payment method, status, and associated task.



UNIVERSITY OF CALOOCAN CITY

Computer Studies Department

Bachelor of Science in Information System



User Interface and Experience

- To design a responsive and user-friendly interface that can be accessed through desktop and mobile web browsers.
- To organize task information using clear visual hierarchy and accessible navigation.
- To provide intuitive forms for creating, editing, requesting, and managing tasks.
- To provide clear validation and error messages when users enter invalid or incomplete information.
- To provide confirmation messages for important actions such as accepting, cancelling, completing, or reporting a task.
- To provide separate dashboards appropriate to the permissions and responsibilities of each user role.
- To provide clear forms and information displays for recording and viewing payment details associated with completed tasks.

Scope: (per User)

TaskTap will have a role-based access model: Admin, Standard User and Guest users. The Admin will have the most system access and will be responsible for managing users, tasks, reports and other system records. Standard Users will be able to post tasks and provide task help, and Guest Users will be able to browse the publicly accessible task info with limited features.

For TaskTap's Admin:

- User management. The Admin can view, manage, archive, and monitor registered user accounts.
- Task management. The Admin can view and manage task postings within the system.
- Category management. The Admin can create, edit, and archive task categories.
- Report management. The Admin can review reports submitted by users regarding tasks or other users.
- Account moderation. The Admin can take appropriate administrative action on accounts that violate system rules.
- Record management. The Admin can archive appropriate system records through soft deletion.
- System monitoring. The Admin can view relevant task, user, and activity information through an administrative dashboard.



UNIVERSITY OF CALOOCAN CITY

Computer Studies Department

Bachelor of Science in Information System



- Payment record management. The Admin can view payment records associated with completed tasks for monitoring and administrative purposes.

For TaskTap's Standard User:

- User authentication. The Standard User can register, log in, log out, and manage their account.
- Profile management. The Standard User can create and update their profile, skills, and other permitted information.
- Task posting. The Standard User can create, edit, and manage task postings.
- Task browsing. The Standard User can browse available tasks based on location, category, price, and deadline.
- Task searching and filtering. The Standard User can search and filter available tasks.
- Task requests. The Standard User can request available tasks that they are interested in completing.
- Request management. A task poster can review and accept or reject requests from other users.
- Task management. Users can update applicable task statuses as the task progresses.
- Messaging. Users involved in a task can communicate through the application's messaging feature.
- Ratings and reviews. Users can rate and review each other after a completed task.
- Notifications. Users can receive notifications related to their tasks, requests, payment updates, messages, and other relevant activities.
- Reporting. Users can report inappropriate tasks or other users.
- Cancellation. Users can cancel applicable tasks or requests according to the system's rules.
- Map viewing. Users can view approximate task locations through the system's map interface.
- Payment status. The Standard User can view the current payment status associated with their applicable tasks.
- Payment recording. The Standard User can record and view payment information associated with applicable tasks, including the agreed amount, payment method, payment status, and payment date.

For TaskTap's Guest User:

- Minimal browsing. The Guest can browse publicly available task listings.



UNIVERSITY OF CALOOCAN CITY

Computer Studies Department

Bachelor of Science in Information System



- Task searching. The Guest can search and filter publicly available task information.
- Task information. The Guest can view basic information about available tasks.
- Registration. The Guest can access the registration process to become a Standard User.
- Authentication. The Guest must register and log in before accessing features that require an authenticated account.

•

Limitations:

- The system will not provide live GPS tracking or continuous real-time location sharing between users.
- The system will not handle actual online payment processing
- The map feature will display approximate task locations and will not reveal a user's precise real-time location.
- The system will not guarantee the quality, safety, or completion of tasks performed by users.
- The system will not automatically match users with tasks using an advanced recommendation or artificial intelligence system.
- The system will not provide background checks or external criminal-record verification for users.
- The system will not function as a full employment or payroll management system.
- The availability and accuracy of third-party map services may affect the map functionality of the system.
- The system will not automatically transfer, distribute, or disburse payments to task helpers.
- The messaging feature will not support real-time communication. Messages will be stored in the system and retrieved when users access the conversation.

Proponents:

1. Pineda, Joshua D.
2. Dreza, Sean Matthew
3. Junio, Zef Ian B.
4. Terencio, Jovan Flويدazamie P.

Proposed to:

Prof. Romeo Raphael John Q. Flor

Subject Professor, CCS 116 – Web Development 2